KALI LINUX NETWORK CONFIGURATION

Using VirtualBox

1. Project Overview

This project documents the installation of 7-Zip and Oracle VirtualBox, followed by manual IPv4 network configuration and connectivity verification in Kali Linux. A custom NAT Network is configured with the 10.0.0.0/24 subnet, and Kali Linux is assigned the IP address 10.0.0.2/24.

2. Objectives

- Install 7-Zip and Oracle VirtualBox on a Windows host.
- Import and configure a Kali Linux virtual machine.
- Create or configure a NAT Network using 10.0.0.0/24.
- Set a manual IPv4 address on Kali Linux.
- Restart the network connection and verify the configuration.
- Test connectivity and prepare the lab for authorized cybersecurity practicals.

3. Requirements

- Windows host operating system
- Oracle VirtualBox
- Kali Linux virtual machine
- 7-Zip
- Internet connection

4. Tools and Resources

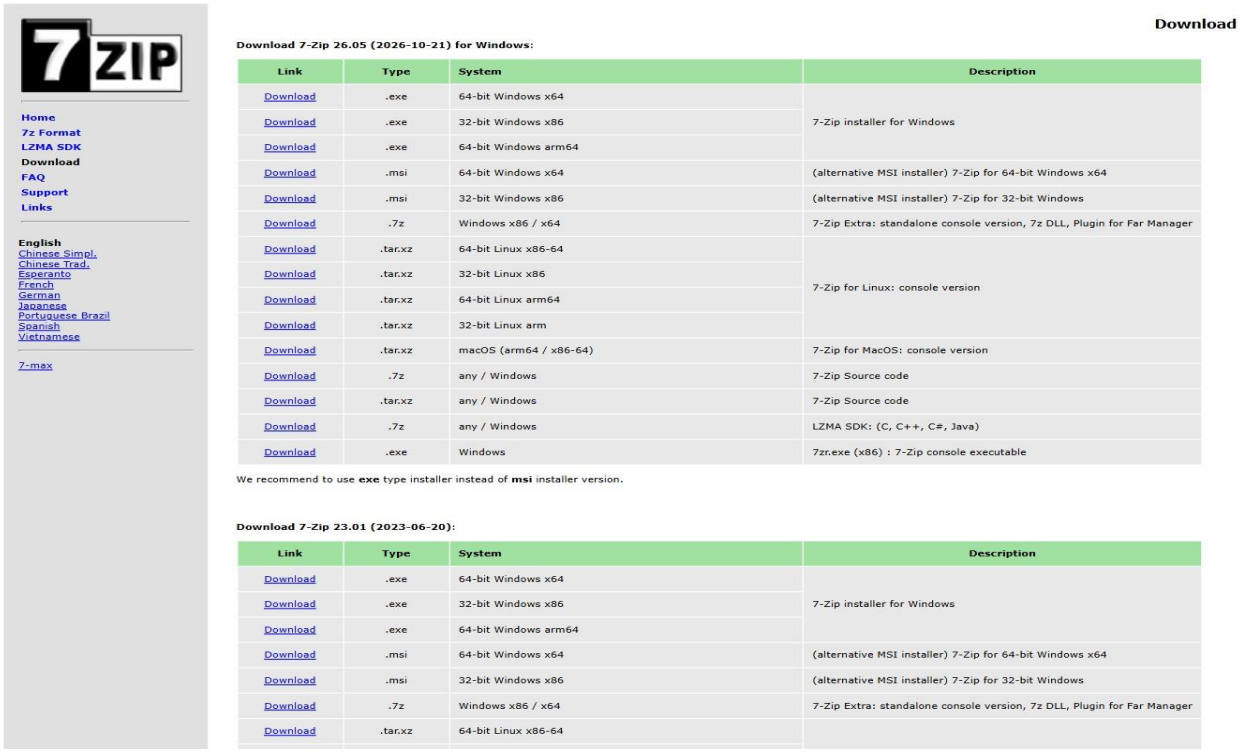
Tool / Resource	Purpose	Link
7-Zip	Extract compressed files when required.	https://www.7-zip.org/
Oracle VirtualBox	Run virtual machines.	https://www.virtualbox.org/wiki/Downloads
Kali Linux	Linux environment for cybersecurity lab practice.	https://www.kali.org/get-kali/
VirtualBox Manual	Reference for network and VM settings.	https://www.virtualbox.org/manual/
Kali Linux Documentation	Reference for Kali configuration.	https://www.kali.org/docs/

5. Lab Configuration

Parameter	Value
Host Operating System	Windows
Virtualization Software	Oracle VirtualBox
Virtual Machine	Kali Linux
Network Mode	NAT Network
IPv4 Network Prefix	10.0.0.0/24
Kali IPv4 Address	10.0.0.2
Prefix Length	24
Gateway	10.0.0.1
DNS Server	8.8.8.8

6. Lab Setup Procedure

Step 1: Install 7-Zip



7-Zip

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7-max

Download

Download 7-Zip 26.05 (2026-10-21) for Windows:

Link	Type	System	Description
Download	.exe	64-bit Windows x64	7-Zip installer for Windows
Download	.exe	32-bit Windows x86	
Download	.exe	64-bit Windows arm64	
Download	.msi	64-bit Windows x64	(alternative MSI installer) 7-Zip for 64-bit Windows x64
Download	.msi	32-bit Windows x86	(alternative MSI installer) 7-Zip for 32-bit Windows
Download	.7z	Windows x86 / x64	7-Zip Extra: standalone console version, 7z DLL, Plugin for Far Manager
Download	.tar.xz	64-bit Linux x86-64	7-Zip for Linux: console version
Download	.tar.xz	32-bit Linux x86	
Download	.tar.xz	64-bit Linux arm64	
Download	.tar.xz	32-bit Linux arm	
Download	.tar.xz	macOS (arm64 / x86-64)	7-Zip for MacOS: console version
Download	.7z	any / Windows	7-Zip Source code
Download	.tar.xz	any / Windows	7-Zip Source code
Download	.7z	any / Windows	LZMA SDK: (C, C++, C#, Java)
Download	.exe	Windows	7zr.exe (x86) : 7-Zip console executable

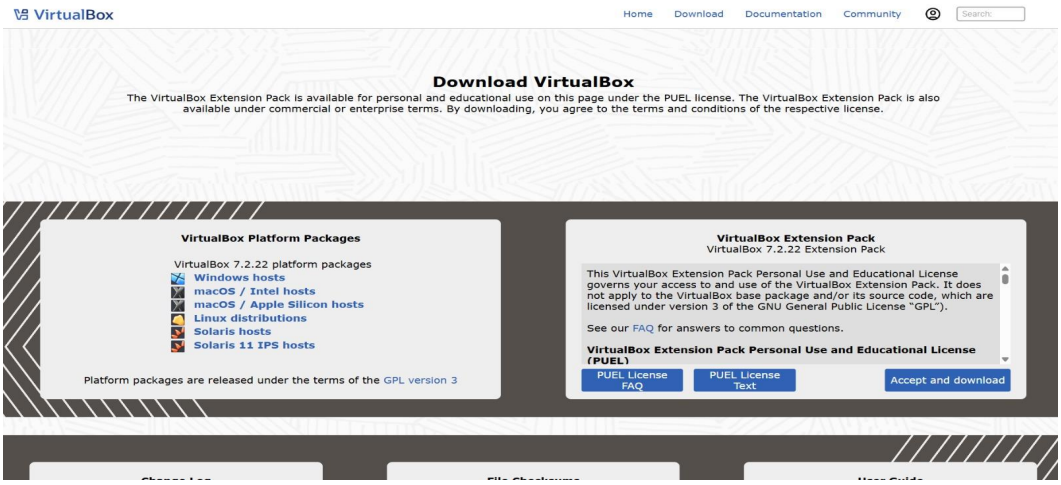
We recommend to use **exe** type installer instead of **msi** installer version.

Download 7-Zip 23.01 (2023-06-20):

Link	Type	System	Description
Download	.exe	64-bit Windows x64	7-Zip installer for Windows
Download	.exe	32-bit Windows x86	
Download	.exe	64-bit Windows arm64	
Download	.msi	64-bit Windows x64	(alternative MSI installer) 7-Zip for 64-bit Windows x64
Download	.msi	32-bit Windows x86	(alternative MSI installer) 7-Zip for 32-bit Windows
Download	.7z	Windows x86 / x64	7-Zip Extra: standalone console version, 7z DLL, Plugin for Far Manager
Download	.tar.xz	64-bit Linux x86-64	

1. Visit https://www.7-zip.org/
2. Download the appropriate Windows installer.
3. Run the installer and complete the installation.

Step 2: Install VirtualBox



VirtualBox

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Solaris hosts
Solaris 11 IPS hosts

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VirtualBox 7.2.22 Extension Pack

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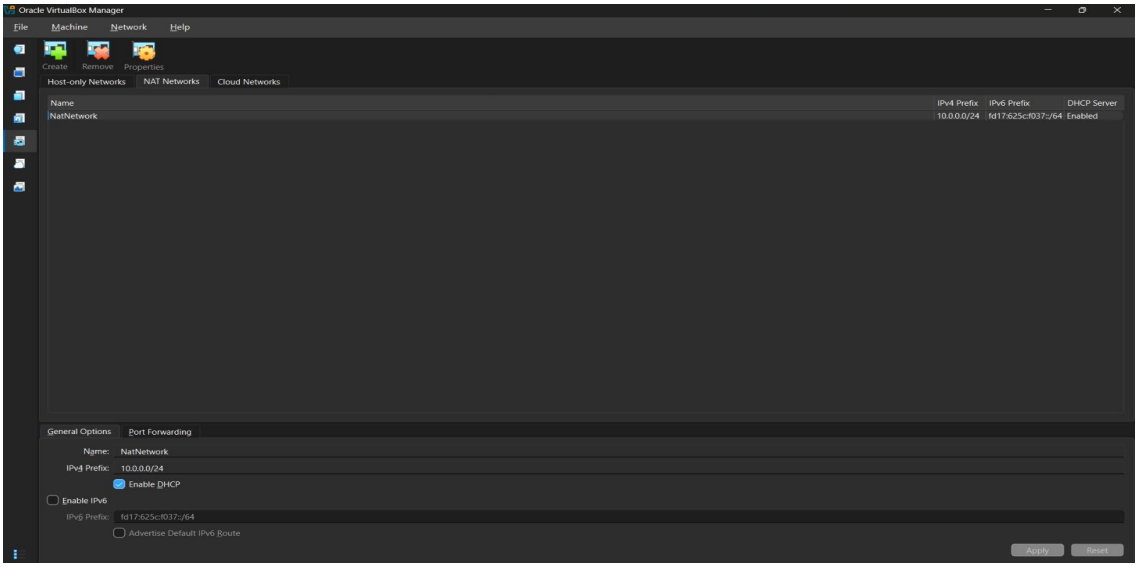
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1. Visit https://www.virtualbox.org/wiki/Downloads
2. Download the Windows hosts package.

3. Run the installer and complete the setup.

4. Open VirtualBox Manager.

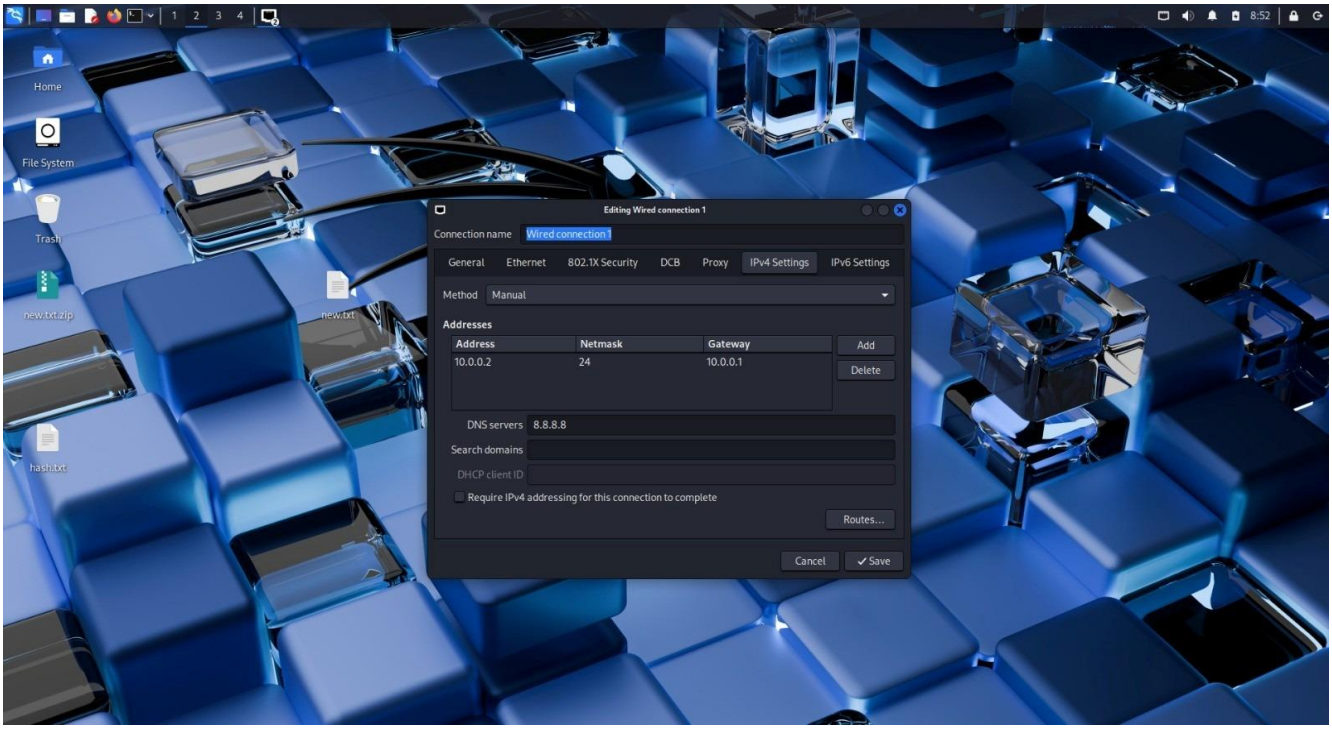
Step 3: Configure the NAT Network



1. Open VirtualBox Manager.
2. Navigate to Network and select NAT Networks (the menu location may differ by VirtualBox version).
3. Create or select NatNetwork.
4. Set the IPv4 prefix to 10.0.0.0/24.
5. Enable DHCP if required by the lab configuration.
6. Save the settings.

Step 4: Configure Kali Linux IPv4

Open the network settings and edit "Wired connection 1". Configure the following values:

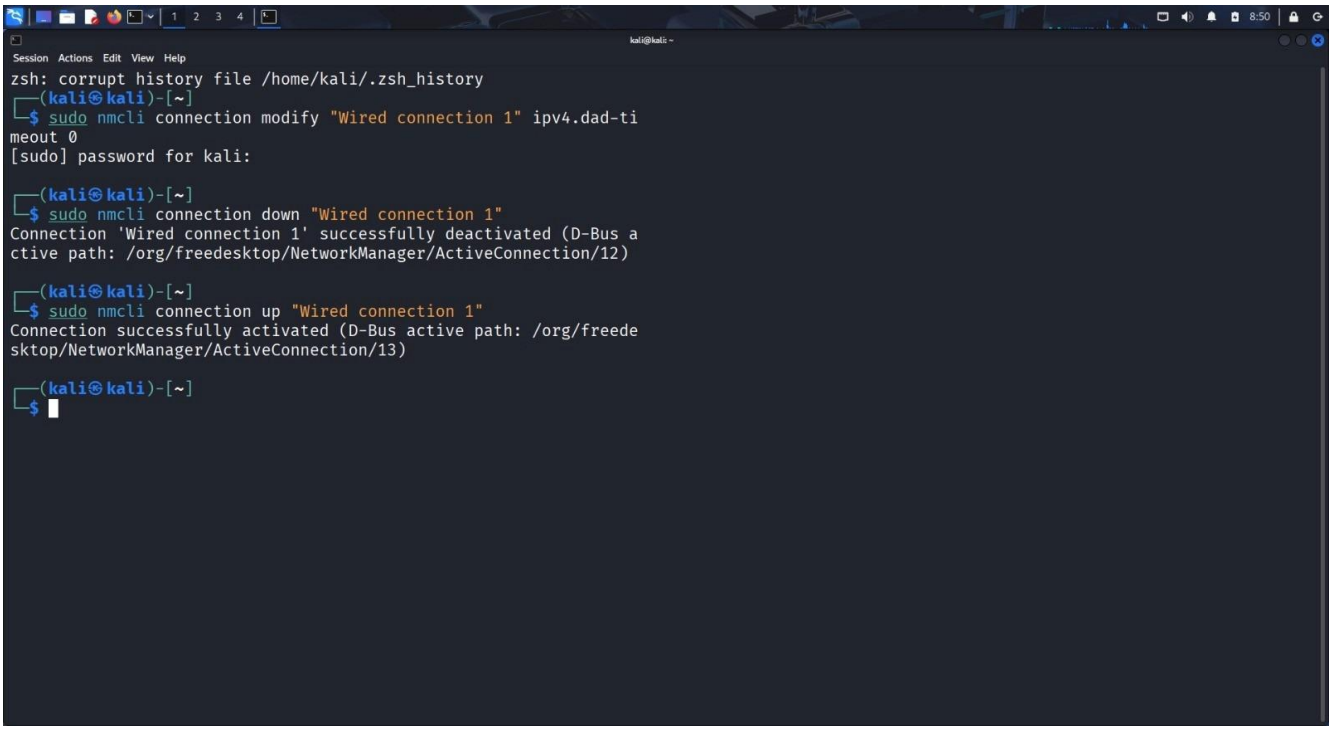


Parameter	Value
Method	Manual
IP Address	10.0.0.2
Prefix	24
Gateway	10.0.0.1
DNS Server	8.8.8.8

Click Save.

Step 5: Restart the Connection

Run these commands in the Kali Linux terminal:




```
sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0
sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```

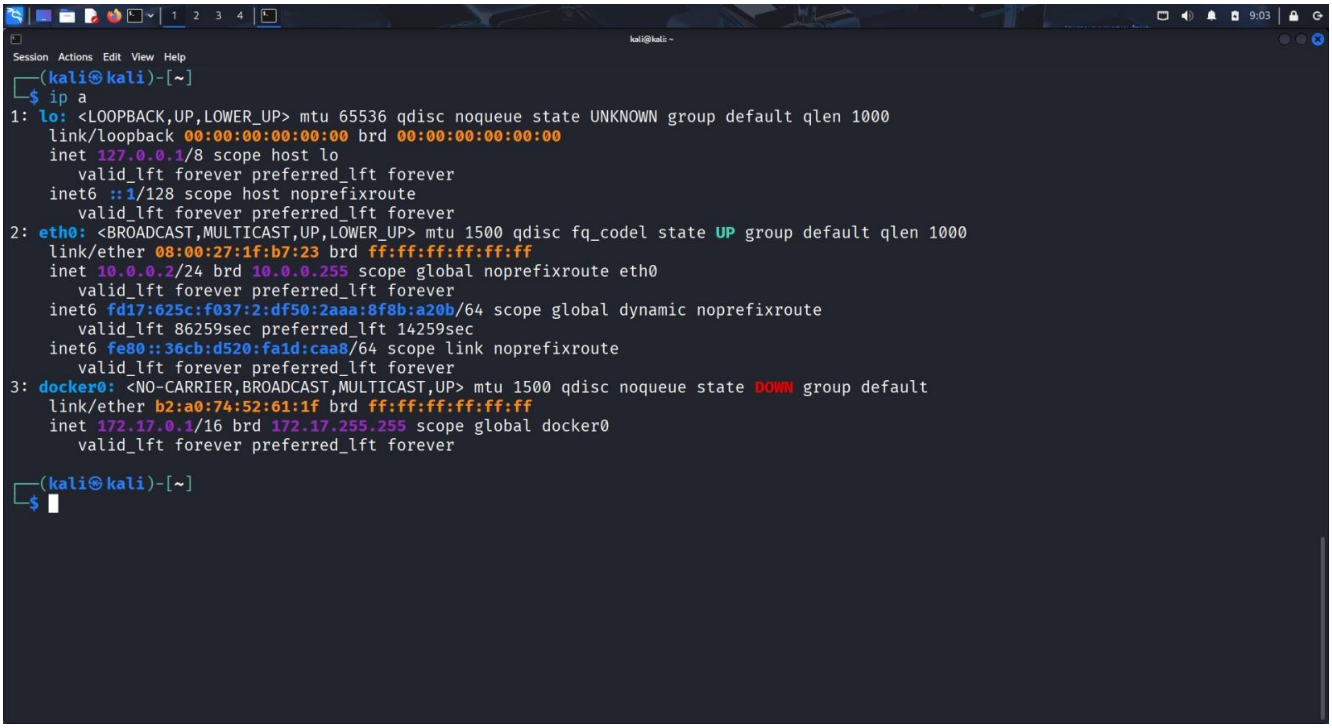
Step 6: Verify the Network Configuration

Execute:

```
ip a
```

Verify that the relevant network interface displays the expected IPv4 address:

```
inet 10.0.0.2/24
```



7. References

- 7-Zip: https://www.7-zip.org/
- VirtualBox Downloads: https://www.virtualbox.org/wiki/Downloads
- Kali Linux: https://www.kali.org/get-kali/
- VirtualBox Manual: https://www.virtualbox.org/manual/
- Kali Linux Documentation: https://www.kali.org/docs/